BrandLens

Check content against the brand's own approved facts and rules
before it goes public, and explain every finding.

Round 1 - The Plan | Problem statement by Inkloom AI | Team Achivers | R S AJAAY

1. Understanding of the Problem

Organisations now publish through employees, agencies, freelancers and AI tools. The brand rules sit in PDFs and in people's heads, so a wrong claim, an outdated price, a banned term or a missing image description often reaches the public before anyone notices. Reviewers catch these by hand if at all, and a rejection that says only "off-brand" teaches the creator nothing.

- **Terminology and tone drift:** one product, three names.
- **Outdated or wrong facts:** old prices, discontinued offers.
- **Unsupported promotional claims:** "best in class" with no approved evidence behind it.
- **Visual and accessibility gaps:** wrong logos, low contrast, missing alt text.
- **Unexplained rejections:** no named rule, so nothing to learn from and nothing to audit.

The brief does not ask for a stricter gatekeeper. It asks for a reviewer that works from the organisation's own approved material, explains each finding, suggests fixes that keep the creator's purpose, and leaves the decision to people. It also asks for insight into repeated mistakes and unclear guidelines.

The part I will solve

The **pre-publication review loop**: approved knowledge base, then cited findings and suggested corrections, then human approval, then recurring-issue insights. Text first, one image format later. Out of scope for this event: a full enterprise compliance suite, direct CMS publishing and video review.

Problem validation (current status)

My understanding so far comes from the event brief. Independent user validation is not done yet, so the needs described here are hypotheses. Planned before Round 2: a short survey (8-10 questions, target 15 or more responses) and 3-5 interviews with people who write or review content. Results will be reported with sample size and limitations, including anything that contradicts my assumptions. The demo brand is fictional test data, not evidence.

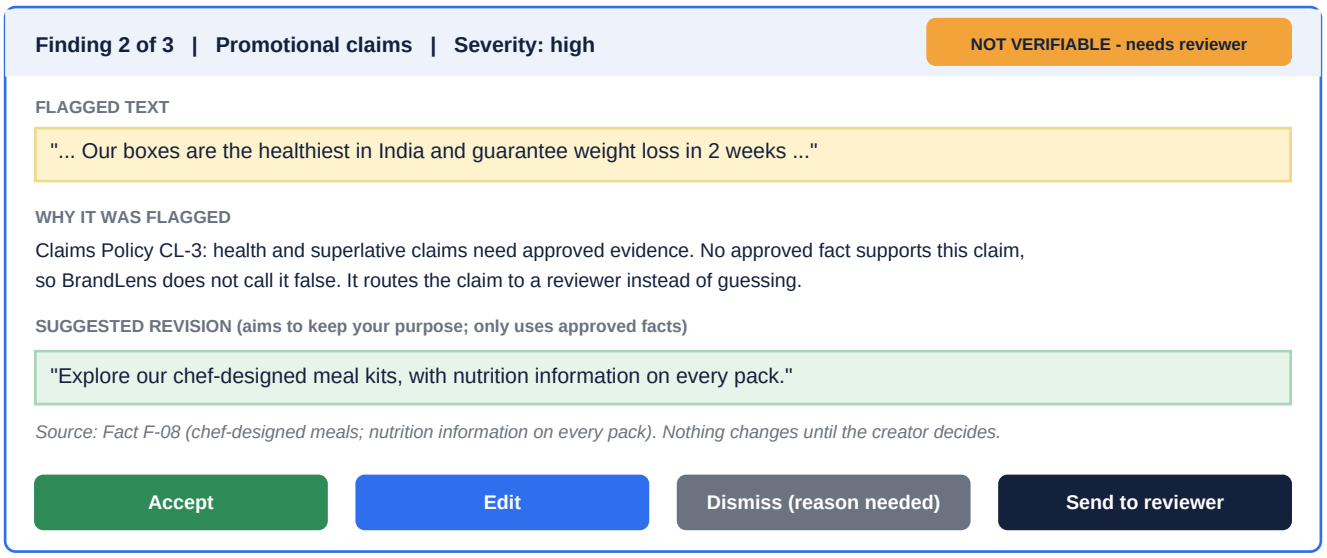
2. Users Affected

User	Pain today	What BrandLens gives them
Content creator (employee, agency writer, designer)	Cannot tell if wording, claims or visuals comply; feedback arrives late	Instant feedback with the reason, the rule and a fix
Brand or marketing manager	Reviews everything by hand; guidelines are not enforced	One place for approved facts and terms; recurring-issue view
Reviewer (legal, compliance, product)	Has to hunt for evidence behind a claim	The claim, its evidence and the rule on one screen; audit trail
Admin / security owner	Confidential brand material shared loosely	Role-based access, workspace isolation, AI-use disclosure

Setting: a mid-size company or agency with several contributors, a guideline document, a fact sheet and an approval chain, where review currently happens over documents, chat and email.

3. Proposed Solution and Value Proposition

BrandLens is a pre-publication review platform. A brand manager loads guidelines, approved facts, terminology and assets into a private knowledge base. A creator submits content; BrandLens checks it against that material and returns findings. Every finding names the rule or fact behind it and offers a correction. People accept, edit or dismiss it, and reviewers approve through a role-based workflow with comments and revision history.



Mockup of one finding (fictional brand FreshNest). Its two siblings in the same review: "diet boxes" breaks Glossary T-07 (use "meal kits"), and "free delivery on all orders" contradicts Fact F-12 (free delivery on orders of Rs 499 and above).

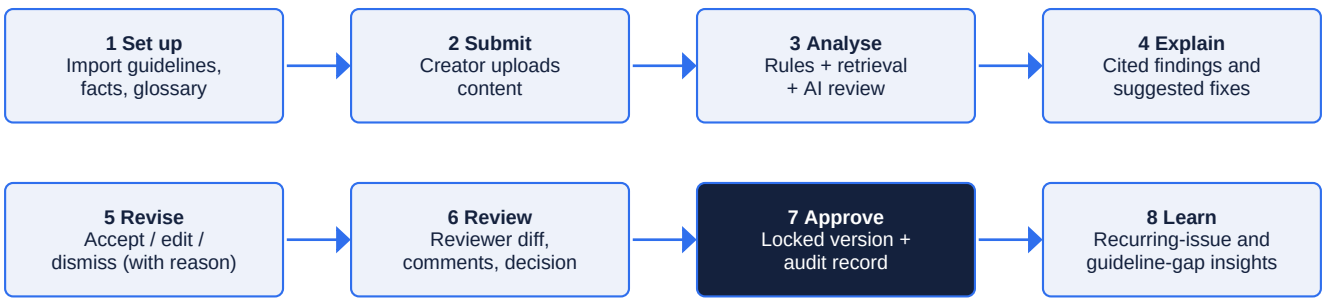
Three kinds of finding

Type	Meaning	What the system does
Rule violation	Content breaks a stated guideline or glossary rule	Cites the rule ID; suggests the compliant wording
Contradicted claim	Content conflicts with an approved fact	Cites the fact ID; suggests the correct statement
Not verifiable	No approved evidence was found for the claim	Does not call it false; routes it to a reviewer with the relevant policy

What makes this approach different

- **"Not verifiable" is a real outcome,** so the tool does not guess when evidence is missing.
- **Corrections are constrained to approved facts.** A suggestion that adds an unsupported fact is rejected or sent to a human.
- **Disputed rules are treated as feedback on the guidelines,** not only as user error.
- **Deterministic checks first, AI second,** with accuracy measured on a labelled test set.

4. Complete Product Workflow



Step	What happens
1 Set up	Manager imports guidelines, glossary and approved facts, split into entries with an ID and version, and approves them before use.
2 Submit	Creator pastes or uploads content and picks the content type. Round 2 handles text; Round 3 adds PNG/JPG.
3 Analyse	Banned terms and required phrasing are checked by code. Retrieval pulls the relevant approved entries and an AI model checks tone and claims against them. Contrast and alt-text checks run only when the submission includes colour pairs or an alt-text field.
4 Explain	Each finding shows type, severity, the excerpt, the cited source, the reason and a suggested rewrite.
5 Revise	Creator accepts, edits or dismisses (a dismissal needs a reason) and resubmits. Every version is kept.
6 Review	Reviewer sees the diff, open findings and comments, then approves, rejects or requests changes.
7 Approve	The version is locked with an audit record: who approved, which guideline versions applied, whether an AI service was used.
8 Learn	The dashboard shows recurring issues by rule and content type. Rules that are often dismissed or disputed may indicate unclear guidelines.

5. Main Features

Committed for the final build	Stretch goals (if time allows)
Knowledge base for guidelines, glossary and approved facts Text review with the three finding types and cited sources Suggested edits; accept, edit, dismiss Roles, comments, approval flow, revision history Basic recurring-issue dashboard PNG/JPG submission with an alt-text field and contrast checks for user-specified colour pairs (not a full accessibility audit)	Logo misuse detection General visual-style analysis PDF and slide import Advanced confidence scoring Stale-fact alerts

The three improvements I name for judging

- **1. Citation-backed explanations:** every finding links to a validated source and is one of the three types.
- **2. Guideline-gap insights:** a dashboard of recurring mistakes and frequently disputed rules.
- **3. Intent-preserving rewrites with a side-by-side diff:** suggestions aim to keep the creator's meaning, the creator confirms the wording, and nothing is applied automatically.

6. Technology Stack and Architecture

Layer	Choice	Why
Frontend	React, Tailwind CSS	Review screens, diff view, dashboard
Backend	Python, FastAPI	Python is what I already use for ML work; quick APIs
Database	PostgreSQL with pgvector	Users, versions and comments plus retrieval in one system
AI layer	Hosted LLM API with schema-validated JSON output; embeddings	Findings must be tied to retrieved approved sources
Rule engine	Deterministic checks (glossary, banned terms, WCAG contrast formula)	Predictable and explainable for hard rules

Layer	Choice	Why
Auth, hosting	JWT roles, encrypted storage, Docker, free-tier cloud, GitHub	Controlled access; judge-accessible demo and repository

Hardest technical challenge

Telling a real brand violation from an acceptable creative variation without inventing rules. BrandLens retrieves the relevant approved sources and validates every cited source: it must exist, belong to the right workspace, be an approved version and match the quoted text. A valid source ID alone is not proof, so the evaluation separately checks whether the cited passage actually supports the finding. Anything unsupported is labelled needs-human-review.

How reliability will be tested

- A labelled set of about 100 short samples for the fictional brand: compliant text plus text with seeded errors (terminology, contradicted facts, unsupported claims, tone).
- Report issues caught (found / seeded), false-positive rate (compliant samples wrongly flagged / all compliant samples), whether citations support their findings, and the share of suggestions testers accept, failures included.
- Compare reviewer time per item with and without BrandLens in a small user test.

Data: organisation-provided guidelines and facts; for the demo, a fictional brand with its own guidelines, fact sheet and sample content. Every API, model, library and dataset used will be disclosed in Round 2.

7. Team Responsibilities

This is a **solo entry**: I own research, design, backend and AI pipeline, frontend, testing and deployment. To keep that realistic I am building a text-first MVP, using managed services instead of my own infrastructure, and finishing one complete flow before any extension.

Why I can build this

I built a TB detection tool from chest X-rays in Python: an ML model that classifies images as tuberculosis-positive or normal, Grad-CAM heatmaps that highlight the regions behind each prediction, and a web interface where a user uploads an image and gets back the classified image with a confidence score and marked suspected regions. BrandLens follows the same principle: an automated judgement is only useful if a person can see what it is based on, how sure it is, and can overrule it. Here the heatmap becomes a cited rule or fact, and the confidence score becomes the "needs human review" state.

8. Development Plan Across Three Rounds

Round	Goal	Deliverables
1 - The Plan	Define problem, users, solution, architecture	This plan; survey and interview design; outline of the fictional demo brand
2 - The Build	Core workflow runs end to end	Knowledge base import; text review with three finding types and cited sources; suggestions; roles, approval flow, comments, revision history; deployed demo and public repository; architecture note; tools and AI disclosed; survey results; first user feedback
3 - The Finish	Reliable, polished product	Recurring-issue dashboard; PNG/JPG review with alt-text field and contrast checks for user-specified colour pairs; accuracy results on the test set; fixes made; privacy, security and accessibility note; demo video; final report with limitations and roadmap

Main risks

Risk	Mitigation
Model invents a rule or misquotes a source	Source validation in code; unsupported findings become needs-human-review
False positives frustrate creators	Deterministic rules where possible; severity levels; dismissal reasons feed the insights
Confidential data reaches an external AI service	Minimal excerpts; provider terms checked and disclosed first; external-AI-off mode
Scope too big for one person	Committed vs stretch list; one end-to-end flow first

9. Expected Impact

- **Organisations:** fewer public brand errors, fewer unsupported claims, shorter review cycles.
- **Creators:** immediate feedback that explains the rule instead of just blocking work.
- **Reviewers and brand teams:** time moves from proofreading to judgement; every approval is traceable; unclear guidelines become visible.
- **Measured by:** detection and citation accuracy on the test set, suggestion acceptance, and reviewer time per item.

10. Future Scope

- Review inside tools people already use: CMS, Google Workspace, Slack, design tools.
- More formats (slides, HTML, video captions) and languages; detection of conflicts between guideline entries.
- Self-hosted AI option for very confidential organisations; multi-brand workspaces and industry compliance packs.

Responsible Design

- **Privacy:** data isolated per organisation, encrypted storage, role-based access, minimal excerpts sent to external services. BrandLens will not train its own models on customer content; external providers' training, retention and processing terms will be checked and disclosed before confidential data is used.
- **External-AI-off mode:** disables all external model calls, including hosted embedding services. Users keep deterministic terminology and required-phrase checks, manual review, comments and approvals; AI tone analysis and suggested rewrites are unavailable unless a local model is configured.
- **Human oversight:** no auto-publishing or auto-rewriting; approvals always by a person; dismissals logged with reasons.
- **Accessibility:** checks on submitted content, and a keyboard-friendly, readable review interface.

11. Team and Participant Details

Field	Details
Team name	Achivers
Participant (sole member and team leader)	R S AJAAY
Role	Product design, research, backend and AI, frontend, testing and documentation (solo)
Individual contribution	Stated separately in the Round 1 submission form
Tools and AI disclosure	Claude (Anthropic) and Perplexity helped draft and review this plan. I made the scope and design decisions and am responsible for its content. APIs, models and libraries will be listed in Round 2.